StudySapiens Board MCQs

Chapter 10 — Human Respiratory System

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. External respiration refers to:

(a) Only photosynthesis
(b) Only cellular respiration
(c) Only glycolysis
(d) Uptake of oxygen and disposal of carbon dioxide at body-system level

2. The actual site of gas exchange in humans is the:

(a) Trachea
(b) Bronchi
(c) Nasal cavity
(d) Alveoli

3. Which of the following is NOT a property of an efficient respiratory surface?

(a) Moist
(b) Thick and dry
(c) Thin
(d) Rich blood supply

4. The respiratory surface is located internally mainly to:

(a) Increase thickness
(b) Increase water loss
(c) Reduce blood supply
(d) Prevent water loss

5. The upper respiratory tract includes:

(a) Lungs only
(b) Alveoli only
(c) Trachea and bronchi only
(d) Nasal cavity, pharynx and larynx

6. The lower respiratory tract includes:

(a) Nose and pharynx only
(b) Larynx only
(c) Trachea, bronchi, bronchioles and lungs
(d) Nasal cavity only

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7. The nasal cavity warms and humidifies air using:

(a) Pleura
(b) Alveoli
(c) Blood vessels and mucous membranes
(d) Cartilage only

8. The voice box is the:

(a) Larynx
(b) Trachea
(c) Bronchus
(d) Pharynx

9. The epiglottis prevents food from entering the:

(a) Stomach
(b) Oesophagus
(c) Pharynx only
(d) Larynx/trachea

10. The trachea is kept open by:

(a) Smooth muscle only
(b) C-shaped cartilage rings
(c) Elastic fibres only
(d) Bone rings

11. The trachea divides into:

(a) Four bronchioles directly
(b) Two primary bronchi
(c) Alveoli directly
(d) Three primary bronchi

12. Bronchioles lack:

(a) Blood supply
(b) Smooth muscle
(c) Epithelium
(d) Cartilage plates

13. The functional units of the lungs are:

(a) Alveoli
(b) Trachea
(c) Bronchi
(d) Pleura

14. The membrane covering the lungs is the:

(a) Peritoneum
(b) Pericardium
(c) Parietal pleura
(d) Visceral pleura

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15. The pleural cavity contains:

(a) Pleural fluid
(b) Blood only
(c) Air only
(d) Lymph only

16. The right lung has:

(a) One lobe
(b) Two lobes
(c) Three lobes
(d) Four lobes

17. The left lung is smaller to accommodate the:

(a) Spleen
(b) Heart
(c) Liver
(d) Stomach

18. Surfactant in alveoli:

(a) Transports oxygen
(b) Digests proteins
(c) Reduces surface tension and prevents collapse
(d) Increases surface tension

19. Inspiration occurs when the diaphragm:

(a) Moves upward only
(b) Contracts and flattens
(c) Relaxes and domes upward
(d) Stops moving

20. During inspiration, the intercostal muscles:

(a) Contract and raise ribs
(b) Relax and lower ribs
(c) Contract and lower ribs
(d) Have no role

21. During inspiration, pressure inside the lungs becomes:

(a) Higher than atmospheric pressure
(b) Zero
(c) Equal to atmospheric pressure always
(d) Lower than atmospheric pressure

22. Expiration is mainly caused by:

(a) Contraction of diaphragm only
(b) Active pumping by alveoli
(c) Increase in thoracic volume
(d) Relaxation of diaphragm and intercostal muscles

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23. Breathing is harder at high altitudes because:

(a) CO_2 is absent
(b) Oxygen percentage is much higher
(c) Air is warmer only
(d) Atmospheric pressure is lower

24. Most oxygen in blood is transported:

(a) Dissolved in plasma only
(b) As bicarbonate
(c) Bound to haemoglobin
(d) Bound to plasma proteins only

25. Oxyhaemoglobin is:

(a) Dark red
(b) Blue
(c) Bright red
(d) Colourless

26. Deoxyhaemoglobin is:

(a) Green
(b) Yellow
(c) Bright red
(d) Dark red

27. At alveolar pO_2 of about 105 mm Hg, haemoglobin saturation is about:

(a) 50%
(b) 35%
(c) 75%
(d) 97%

28. In resting tissues, venous pO_2 is about:

(a) 80 mm Hg
(b) 40 mm Hg
(c) 105 mm Hg
(d) 20 mm Hg

29. The Bohr effect describes the effect of:

(a) Pressure on breathing only
(b) Light on photosynthesis
(c) Temperature on enzymes only
(d) pH/CO_2 on haemoglobin oxygen affinity

30. During exercise, oxygen unloading increases because:

(a) Breathing stops
(b) pCO_2 rises, pH falls and temperature rises
(c) Haemoglobin increases
(d) pO_2 rises in tissues

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31. Approximately 72% of CO_2 is transported as:

(a) Carbonic acid only
(b) Dissolved CO_2 only
(c) Carbaminohaemoglobin
(d) Bicarbonate ions

32. The enzyme carbonic anhydrase catalyses the formation of:

(a) ATP from ADP
(b) Oxyhaemoglobin
(c) Carbonic acid from CO_2 and water
(d) Bicarbonate from glucose

33. The chloride shift involves exchange of bicarbonate ions with:

(a) Potassium ions
(b) Calcium ions
(c) Sodium ions
(d) Chloride ions

34. About 20% of CO_2 is carried as:

(a) Carbaminohaemoglobin
(b) Bicarbonate only
(c) Dissolved CO_2 only
(d) Carbonic acid only

35. CO_2 binds to the _____ of haemoglobin.

(a) Lipid portion
(b) Globin protein portion
(c) Haem iron only
(d) Oxygen-binding site only

36. About 8% of CO_2 is transported:

(a) As bicarbonate only
(b) Dissolved in plasma
(c) As carbon monoxide
(d) Bound to haemoglobin only

37. Carbon monoxide is dangerous because it:

(a) Increases oxygen carriage
(b) Binds haemoglobin more strongly than oxygen
(c) Forms bicarbonate only
(d) Is harmless

38. Haemoglobin contains how many haem groups?

(a) Six
(b) Four
(c) Two
(d) One

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39. Each haem group contains:

(a) An iron ion
(b) A zinc ion only
(c) A calcium ion
(d) A magnesium ion

40. One haemoglobin molecule can carry up to:

(a) Eight oxygen molecules
(b) Two oxygen molecules
(c) One oxygen molecule
(d) Four oxygen molecules

41. Myoglobin is found mainly in:

(a) Red blood cells only
(b) Plasma only
(c) Skeletal and cardiac muscle
(d) Lungs only

42. Myoglobin has:

(a) No haem group
(b) One haem group
(c) Two haem groups
(d) Four haem groups

43. Myoglobin has a _____ affinity for oxygen than haemoglobin.

(a) No
(b) Equal
(c) Higher
(d) Lower

44. Sinusitis is inflammation of the:

(a) Pleura
(b) Sinuses
(c) Alveoli
(d) Bronchioles only

45. Otitis media is inflammation of the:

(a) Sinuses only
(b) Lungs
(c) Larynx only
(d) Middle ear

46. Pneumonia mainly affects the:

(a) Pleura only
(b) Alveoli
(c) Nasal cavity only
(d) Trachea only

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47. A common bacterial cause of pneumonia is:

(a) Streptococcus pneumoniae
(b) Mycobacterium tuberculosis only
(c) HIV
(d) Plasmodium

48. Pulmonary tuberculosis is caused by:

(a) Mycobacterium tuberculosis
(b) Influenza virus only
(c) HIV
(d) Streptococcus pneumoniae

49. Tuberculosis is treated with multiple antibiotics for about:

(a) 1 week
(b) 3 days
(c) 1 month only
(d) 9 months

50. COPD is commonly caused by:

(a) High altitude only
(b) Vitamin deficiency only
(c) Smoking and pollutants
(d) Exercise only

51. Emphysema involves destruction of:

(a) Tracheal cartilage only
(b) Pleural fluid only
(c) Vocal cords only
(d) Alveolar walls

52. A symptom of pulmonary tuberculosis is:

(a) Joint swelling only
(b) Skin rash only
(c) Jaundice only
(d) Cough with blood

53. The respiratory pigment in red blood cells is:

(a) Haemocyanin only
(b) Chlorophyll
(c) Myoglobin only
(d) Haemoglobin

54. The respiratory pigment in muscle cells is:

(a) Haemoglobin only
(b) Chlorophyll
(c) Myoglobin
(d) Melanin

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55. Which condition reduces the oxygen-carrying capacity of blood?

(a) Increased ventilation only
(b) Exercise only
(c) High altitude acclimatization only
(d) Carbon monoxide poisoning

56. The partial pressure of oxygen is highest in the:

(a) Venous blood compared with alveoli
(b) Alveoli/atmosphere compared with tissues
(c) Cells compared with air
(d) Tissues compared with alveoli

57. The partial pressure of carbon dioxide is highest in the:

(a) Alveoli
(b) Arterial blood only
(c) Atmosphere
(d) Tissues

58. The main muscle of breathing is the:

(a) Quadriceps
(b) Tongue only
(c) Diaphragm
(d) Biceps

59. Intercostal muscles are located:

(a) Between ribs
(b) In the heart only
(c) In the abdomen only
(d) In the lungs only

60. The pleural membranes help breathing by:

(a) Digesting food
(b) Filtering blood
(c) Allowing lungs to expand/contract with the thoracic wall
(d) Producing oxygen

61. Which of the following is an upper respiratory tract infection?

(a) Sinusitis
(b) Pneumonia
(c) Tuberculosis only
(d) Emphysema

62. Which of the following is a lower respiratory tract infection?

(a) Sinusitis only
(b) Pneumonia
(c) Otitis media only
(d) Common cold only

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63. A genetic predisposition to COPD may involve deficiency of:

(a) Alpha-1 antitrypsin
(b) Insulin
(c) Thyroxine only
(d) Haemoglobin only

64. The main function of nasal hairs and mucus is to:

(a) Trap foreign particles
(b) Produce oxygen
(c) Transport CO_2 only
(d) Digest food

65. Gas exchange in alveoli occurs by:

(a) Diffusion
(b) Active transport only
(c) Endocytosis
(d) Osmosis only

66. The approximate number of alveoli in adult human lungs is:

(a) Hundreds of millions
(b) A few thousand
(c) A few hundred
(d) Ten

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ANSWER KEY WITH BRIEF REASONS

1. (d) Uptake of oxygen and disposal of carbon dioxide at body-system level — It involves breathing and gas exchange.

2. (d) Alveoli — Alveoli provide the respiratory surface.

3. (b) Thick and dry — Respiratory surfaces must be thin/moist/vascular.

4. (d) Prevent water loss — Internal location protects the moist surface.

5. (d) Nasal cavity, pharynx and larynx — Upper tract conducts/conditions air.

6. (c) Trachea, bronchi, bronchioles and lungs — Lower tract leads to gas-exchange regions.

7. (c) Blood vessels and mucous membranes — Rich blood supply conditions air.

8. (a) Larynx — Larynx contains vocal cords.

9. (d) Larynx/trachea — It covers the glottis during swallowing.

10. (b) C-shaped cartilage rings — Cartilage prevents collapse.

11. (b) Two primary bronchi — Right and left bronchi enter lungs.

12. (d) Cartilage plates — Cartilage disappears in smaller airways.

13. (a) Alveoli — Alveoli perform gas exchange.

14. (d) Visceral pleura — Visceral pleura adheres to lung surface.

15. (a) Pleural fluid — Fluid reduces friction during breathing.

16. (c) Three lobes — Right lung is larger with three lobes.

17. (b) Heart — Cardiac notch makes room for heart.

18. (c) Reduces surface tension and prevents collapse — Surfactant keeps alveoli open.

19. (b) Contracts and flattens — Contraction increases thoracic volume.

20. (a) Contract and raise ribs — They expand the rib cage.

21. (d) Lower than atmospheric pressure — Air flows in down pressure gradient.

22. (d) Relaxation of diaphragm and intercostal muscles — Elastic recoil reduces volume.

23. (d) Atmospheric pressure is lower — Lower pressure reduces air intake gradient.

34. (a) Carbaminohaemoglobin — CO_2 binds globin chains.

35. (b) Globin protein portion — CO_2 binds amino groups of globin.

36. (b) Dissolved in plasma — A small fraction dissolves directly.

37. (b) Binds haemoglobin more strongly than oxygen — CO reduces oxygen transport.

38. (b) Four — Each of four chains has one haem.

39. (a) An iron ion — Fe binds oxygen.

40. (d) Four oxygen molecules — Four haem groups bind four O_2.

41. (c) Skeletal and cardiac muscle — Myoglobin stores oxygen in muscle.

42. (b) One haem group — It is a monomer with one haem.

43. (c) Higher — It stores oxygen tightly.

44. (b) Sinuses — It affects air-filled spaces around nasal cavity.

45. (d) Middle ear — Fluid in Eustachian tube can cause it.

46. (b) Alveoli — Alveoli fill with fluid/pus.

47. (a) Streptococcus pneumoniae — S. pneumoniae is a major cause.

48. (a) Mycobacterium tuberculosis — TB bacillus infects lungs.

49. (d) 9 months — Long combination therapy is required.

50. (c) Smoking and pollutants — Chronic irritation damages lungs.

51. (d) Alveolar walls — Surface area for gas exchange is reduced.

52. (d) Cough with blood — TB causes chronic cough/haemoptysis.

53. (d) Haemoglobin — Hb transports oxygen in blood.

54. (c) Myoglobin — Myoglobin stores oxygen in muscle.

55. (d) Carbon monoxide poisoning — CO occupies haemoglobin.

56. (b) Alveoli/atmosphere compared with tissues — Oxygen diffuses from high to low pO_2.

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24. (c) Bound to haemoglobin — Haemoglobin carries the majority of O2.

25. (c) Bright red — Oxygenated haemoglobin is bright red.

26. (d) Dark red — Reduced haemoglobin is darker.

27. (d) 97% — High alveolar oxygen loads haemoglobin.

28. (b) 40 mm Hg — Tissues have lower oxygen pressure.

29. (d) pH/CO2 on haemoglobin oxygen affinity — Low pH promotes oxygen unloading.

30. (b) pCO2 rises, pH falls and temperature rises — Active tissues promote O2 release.

31. (d) Bicarbonate ions — Bicarbonate is the main transport form.

32. (c) Carbonic acid from CO2 and water — It speeds CO2 hydration.

33. (d) Chloride ions — It maintains electrical neutrality.

57. (d) Tissues — CO2 diffuses from tissues to blood to alveoli.

58. (c) Diaphragm — Diaphragm changes thoracic volume.

59. (a) Between ribs — They move the rib cage.

60. (c) Allowing lungs to expand/contract with the thoracic wall — Pleural linkage couples lung/chest movement.

61. (a) Sinusitis — Sinusitis affects sinuses/upper tract.

62. (b) Pneumonia — Pneumonia affects lungs.

63. (a) Alpha-1 antitrypsin — A1AT deficiency can cause COPD.

64. (a) Trap foreign particles — They clean incoming air.

65. (a) Diffusion — Gases move down partial pressure gradients.

66. (a) Hundreds of millions — About 480 million provide huge surface area.

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